Is the Network Worth Building? LLM Scores vs. LLM-Signed Networks for Measuring Polarization

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Abstract

1 Introduction

1.1 Introduction to the Topic

Polarization, the division of a population into opposing groups that are cohesive internally and antagonistic toward one another, is a general property of networked systems rather than a feature of any single domain. It appears not only in discussions of public policy, but also in scientific communities divided over competing theories, online communities split over products or platforms, organizations whose teams form opposing coalitions, and any setting where actors take sides and interact accordingly. Measuring polarization in a way that works across such settings is therefore a basic problem in social network analysis. To be useful, a measure must also be trackable over time and trustworthy when its value changes.

Existing approaches fall into two broad families. The first reads the *content* of interactions: it infers stance or sentiment from text and aggregates it into a score for a conversation or time window. The second reads the *structure* of interactions: it represents actors as nodes and their interactions as edges, and measures how strongly the resulting graph separates into distinct groups, or, when edges are signed for agreement and disagreement, into opposing ones. The two families have largely developed separately, one looking at what people say and the other at who interacts with whom [11].

Signed networks connect the two. In a signed graph $G = (V, E, \sigma)$, each edge carries a sign $\sigma \in \{+1, -1\}$ (or, in a continuous variant, $\sigma \in [-1, +1]$) encoding affinity or antagonism between its endpoints. Structural balance theory [4] shows that a balanced signed network splits into two internally friendly, mutually hostile factions, which is the ideal case of polarization, and deviations from this ideal can be quantified with spectral and combinatorial scores such as Eigen-Sign and the frustration score [11].

Most real interaction networks, however, are unsigned: replies, mentions, and links record that two actors interacted, not whether the interaction was supportive or hostile. Large language models (LLMs) can now read the text attached to an interaction and assign it a sign, which makes it possible to construct signed networks from raw interaction data at scale.

This yields two ways to turn the same interaction text into a polarization measurement, which we henceforth refer to as **Pipeline A** and **Pipeline B**. In **Pipeline A**, an LLM scores the stance of each interaction and these scores are aggregated directly into a polarization score. In **Pipeline B**, an LLM assigns a sign to each edge, a signed network is built for each time window, and a structural algorithm produces the polarization score from the network. Pipeline A thus follows the first family, with an LLM doing the scoring, and Pipeline B follows the second, with an LLM supplying the signs.

1.2 Motivation

The two pipelines encode different assumptions about where polarization lives. Pipeline A assumes it is visible in what actors say, so if expressed stances are extreme and language is hostile, the group is polarized. Pipeline B assumes it is visible in who interacts with whom, thus polarization is the pattern of alliances and oppositions, regardless of the tone of any single message. These assumptions come apart in realistic cases. Two factions may exchange measured, courteous language while remaining sharply separated in their interaction patterns, which a content-based score would read as low polarization. Conversely, a network may be full of heated exchanges that cut across group boundaries rather than along them. This is antagonism without factional structure, and a structural score should treat it very differently from a content score.

The choice between pipelines also has practical consequences. Building and scoring a signed network adds cost and complexity, including edge construction, per-window graph analysis, and sensitivity to sparsity, so it is worth including only if it captures something that the text alone does not. Pipeline B also depends on the quality of the inferred edge signs, and it is not obvious how much sign accuracy matters for the final polarization score. Finally, a measurement is most valuable when it can anticipate change, so the practical question is not only which score better reflects the present state of a network but also whether the structural view improves forecasts of how polarization will move. Because these questions concern the relationship between content and structure, they are not specific to any one domain, and answers obtained on one kind of network inform how polarization should be measured on others.

1.3 Problem Specification

Given a dataset of timestamped, text-bearing interactions between actors, and an external ground truth for actor group membership that is independent of both pipelines, we compute a polarization score for each time window using Pipeline A and Pipeline B, and ask which score better agrees with the ground truth, under what

conditions, and with what predictive value. The problem is stated independently of domain. Our analysis will follow two main aspects, the *conditions* under which each pipeline is more accurate (window size, graph density, level of polarization), and the *temporal* behavior of the scores, including whether they help predict change in polarization.

1.4 Research Questions and Contributions

This study is guided by the following research questions:

I. **Does routing through a signed network beat direct LLM measurement?** We compare Pipeline A and Pipeline B on the same data and time windows, measuring the correlation of each score with the external ground truth.
II. **Under what conditions does network structure add value?** We hypothesize that Pipeline B wins when polarization is carried by who interacts with whom rather than by what is said, for example when hostility is common but not aligned with group boundaries ("discordant allies"), and that Pipeline A wins in sparse windows where there are too few cycles for structural measures to be reliable. We evaluate this by breaking results down by window size, graph density, and polarization level.
III. **Do more accurate edge signs lead to better polarization measurement?** Within Pipeline B, we ablate sentiment-inferred signs, LLM binary signs, and LLM continuous (confidence-weighted) signs, alongside an unsigned-network baseline. We measure the change in agreement with ground truth as sign quality improves, using ground-truth signs as an upper bound where available.
IV. **Does the structure-aware score add forecasting value beyond the text-only score?** We predict the change in polarization, $\Delta P = P_t - P_{t-1}$, rather than its level, since level prediction mostly rewards copying the last value. We compare a persistence baseline ($\Delta P = 0$), Pipeline A only, Pipeline B only, and a combined model, using rolling-origin out-of-sample error.
V. **(Exploratory) Which factors drive polarization change, and do they differ by pipeline?** Candidate drivers include newcomer share, bridging-user share, structural features, toxicity and scalar intensity, and external events.

In addressing these questions, this paper makes the following contributions:

I. **A controlled head-to-head comparison.** We run a content-first pipeline and a structure-first pipeline on identical data, windows, and ground truth, so that differences in measurement quality can be attributed to pipeline design rather than to the dataset or evaluation.
II. **A characterization of when structure helps.** We identify the regimes (window size, density, polarization level, and the presence of antagonism that crosses group boundaries) in which the signed-network route outperforms direct measurement, and those in which it does not.
III. **An analysis of sign quality.** We quantify how much edge-sign accuracy matters for downstream polarization measurement by comparing sentiment-based, binary LLM, and continuous LLM signs against an unsigned baseline and, where available, ground-truth signs.
IV. **A temporal evaluation.** We test whether the structural score improves forecasts of change in polarization over content-only and persistence baselines, and, as an exploratory analysis, examine which network-level factors accompany that change.

2 Background and Related Work

2.1 Polarization: Definitions and Constructs

At its core, polarization describes a community that splits into opposing camps where members tend to agree and interact positively within their own camp and to clash with members of the other. The phenomenon has been studied most often in political discourse, but the underlying idea applies to any contested topic discussed online. The term, however, is used for many different things, and the literature has not settled on a single definition [23]. Fraxanet et al. observe that it is applied so freely that it lumps together distinct forms of conflict [9]. A systematic review of 94 empirical studies found that roughly two-thirds of the articles gave no definition at all, and many did not say which kind of polarization they measured [18]. Bramson et al. distinguish nine senses of the term that are often conflated in practice [3]. Because our study compares two measurement pipelines, we must state precisely which construct each one captures. We organize the literature into four families of constructs (opinion, affective, discursive, and structural), followed by a process view and the boundaries with neighboring concepts.

Opinion polarization refers to the divergence of opinions, beliefs, attitudes, and stances between opposing sides [18]. It describes how far individuals' positions lie toward opposite ends of a spectrum, whereas issue polarization concerns divergence on a specific policy question [7]. These constructs are usually measured either from placement on an ideological scale or from self-reported stances on a specific topic. Kubin and von Sikorski caution that the two are related through political sorting but are not always comparable, so studies that measure them differently should not be compared directly [18]. For distributions of opinions, statistical bimodality is among the most common quantitative signatures of polarization [3, 7]. These measures can be unreliable, however. In a study of 76 million comments on about 159,000 political YouTube videos, Di Martino et al. show that the Bimodality Coefficient tends to classify skewed but unimodal distributions as bimodal, flagging about 78% of comment sections, whereas the Dip Statistic and their distance-from-unimodality measure classified most sections as unpolarized or weakly polarized [7]. Using an adaptation of burst detection to find modes, they further report that 87% of the 65,928 comment sections with at least 50 labeled users showed a single detected burst, consistent with mostly unpolarized discussions [7]. At the network level, polarization is framed as a macro-level phenomenon in which ideological distances grow across the network, and it is present when the average pairwise distance between connected

users is high [25].

Affective polarization concerns emotions, feelings, and identity rather than policy positions. Grounded in a social identity perspective, it captures the extent to which individuals feel warmth toward their in-group and dislike or hostility toward the out-group [15, 16, 18], a growing emotional distance between factions [7]. In digital environments, it is typically inferred from text through content analysis, sentiment analysis, or moral-emotion words [8]. Chowdhury et al. use LLaMA-3.1-70B to extract stance, affective tone, and agreement between tweets and map these outputs to a score through predefined heuristic rules, which yields an estimate even for a single reply [5].

The third family focuses on how language is used to express and frame a debate. We use **discursive polarization** to refer to how intensely language is used to assert a stance. Guo et al. operationalize this through scalar-intensity language, the degree to which speakers strengthen, exaggerate, or intensify propositions, including intensifiers, evaluative language, emotion terms, and charged descriptors [11]. They treat it as a property of discourse that they relate to structural polarization, rather than as a form of polarization in itself. Closely related is **interpretive polarization**, the process by which opposing groups contextualize the same topic using incompatible vocabularies and frames [8, 17]. Language and network interactions can also move independently. In a study of 1.3 million original COVID-19 vaccine tweets, Durrheim and Schuld found that, among 500 sampled user pairs, distance in the language-based speaker landscape was essentially uncorrelated with shortest-path distance in the retweet network ($r = -0.07$ before the pandemic declaration and $r = 0.10$ after) [8]. The retweet networks stayed relatively stable, whereas the speaker landscapes became polarized between the antivax and public-health poles, with even moderate speakers still sorted by party [8]. This mismatch between what users write and whom they retweet motivates our comparison of text-based and network-based pipelines.

Structural polarization describes how interactions are arranged across a population. It is formalized through structural balance theory where a signed graph is balanced when it can be split into two factions, with positive ties inside each faction and negative ties between them [4]. The degree of imbalance is commonly measured by *frustration*, the number (or weight) of edges that violate the best partition of the graph into factions. Fraxanet et al. build on this idea with the FAULTANA pipeline, which separates two ingredients: *antagonism*, the overall prevalence of negative interactions, and *alignment*, the extent to which those negative interactions fall along a single major dividing line [9]. This separation shows that a topic can trigger intense conflict without aligning with the main division, as they observe for the COVID-19 debate in an online newspaper forum. Guo et al. derive continuous signed edges from LLM stance scores and show that frustration measures normalized by the graph's cycle-space capacity compare networks of different sizes and densities better than raw spectral scores [11]. They also report that scalar-intensity language is associated with structural polarization in the same month, but find little evidence that language in one month predicts polarization in the next [11].

The constructs above are often treated as static states. Treuillier et al. argue instead that polarization is a sequential process. In their study of two online debates, clusters of users evolve through a succession of distinct periods, a group of intermediate users who interact with both sides plays an important role, and context-related events can reset the pattern [23]. Chowdhury et al. similarly find event-dependent dynamics, with polarization rising before well-anticipated events and spiking right after sudden ones [5]. These findings motivate our use of time windows for tracking each pipeline's score.

Polarization is related to, but distinct from, several other phenomena. Echo chambers, filter bubbles, and tunnel vision are studied under the broader heading of ideological isolation and describe how exposure to information and like-minded peers becomes concentrated, whereas polarization is a macro-level division between groups [25]. Extremity also differs from polarization since distribution-based measures detect departure from unimodality, a property of the whole population, rather than the presence of extreme individuals [7]. Finally, hostile language and controversy do not by themselves imply structural polarization, since conflict can arise without aligning with the main dividing line [9, 11].

2.2 From Sentiment Analysis to LLMs for Sign and Stance Inference

Many of the existing approaches that aim to measure affective polarization in networks rely on traditional NLP methods, such as sentiment analysis and stance detection. *Sentiment analysis*, which aims to classify a given text's tone as positive, negative or neutral, often suffers quality degradation when working with complex texts that deal with a lot of implicit meanings, sarcasm, and particularly with identifying the target of a statement with multiple subjects [5, 10]. *Stance detection*, which determines whether the subject producing the text supports, opposes, or is neutral with regards to a specific issue being mentioned, similarly struggles with linguistic nuance and subtle wording [5]. Consequently, LLMs have emerged as a tool capable of offering deep contextual understanding beyond what the traditional methods can offer. This context-aware language processing, owing to the large amount of data they were trained on, makes them uniquely suited for capturing subtle sentiment shifts, varying levels of emotional intensity and rhetorical strategies, and by extension, affective polarization. [5]

2.3 Existing Implementations and Variations

We focus on two ways in which LLMs have been used to measure polarization in online conversations. In the first, LLMs score textual exchanges against predefined criteria such as stance, affective tone, and agreement, and the scores are aggregated into a polarization measure without building a network [5]. In the second, introduced more recently by Guo et al. [11], a stance model converts conversational exchanges into continuous signed agreement and disagreement ties, which form a signed network on which structural polarization is then measured.

Score-based aggregation. Chowdhury et al. [5] use LLMs to extract, from large-scale discussions on climate change and gun control, the stance of a post (support, opposition, or neutrality), the presence of emotionally charged language, and the extent to which replies align or conflict with the stance of the original post. Domain-informed heuristic rules map these attributes to a polarization score between 0 and 10 for each interaction, with low scores reflecting constructive disagreement and high scores reflecting heated conflict and emotional hostility. For longer conversations, the score is the average over individual interactions, and polarization trends are then evaluated over time. The authors present this as an alternative to sentiment analysis, stance detection, and network-based measures, and argue that, unlike network-based measures, their score is explainable and can be computed for conversations as small as a single interaction.

Signed-network construction. Guo et al. [11] instead use language to define the network itself. They adapt a pretrained language to a three-class stance task (disagree, neutral, agree), trained on the a human-annotated dataset, and combine the class probabilities into a continuous edge weight $w = (p_{\text{agree}} - p_{\text{disagree}})(1 - p_{\text{neutral}}) \in [-1, 1]$. Neutral or uncertain exchanges are thus shrunk toward zero rather than treated as full-strength ties. The authors motivate this by noting that hard labels discard model confidence, and that because signed polarization measures are highly sensitive to edge signs, this injects label noise directly into the structural estimate. Interactions are grouped into monthly windows, each yielding a signed interaction graph whose structural polarization is computed.

Variations. The two approaches differ in what the LLM outputs (categorical attributes combined by rules versus class probabilities turned into a continuous weight) and in how outputs are aggregated (averaging over interactions versus partitioning a signed graph). The second choice carries a structural caveat. Guo et al. [11] note that structural polarization scores depend on the size and density of the graph, so raw values can drift across time windows or networks even when the underlying signed structure is unchanged, and they normalize their scores to make windows comparable. Their synthetic benchmarks add that different structural indices recover similar two-camp partitions in dense, high-signal graphs but diverge in sparse ones. Structural indices are therefore not interchangeable, and their behavior depends on the shape of the network they are applied to.

In this study, we use both approaches as the basis for **Pipeline A** and **Pipeline B** respectively.

2.4 Temporal Dynamics of Polarization

Polarization is not a fixed property of a community but a process that unfolds over time. A single snapshot can show how divided a group is at one moment, but it cannot show how that division formed, whether it is deepening or fading, or what triggered the change. Treating polarization as an evolving quantity shifts the question from *how polarized is this network?* to *how does polarization change, and why?* Formally, if G_t denotes the interaction network observed in time window t, and $P(G_t)$ is a polarization score computed on it, the object of study becomes the sequence $\{P(G_1), P(G_2), \ldots, P(G_T)\}$ rather than any single value.

Empirical work suggests that polarization often develops through recognizable stages rather than appearing all at once. A newly formed discussion typically has little group structure. Distinct camps then emerge, often alongside a central group of participants who have not committed to either side. Over time these intermediate participants tend to drift toward one of the poles, and once they disappear the community settles into a rigid two-sided configuration [23]. This implies that the same score can mean different things at different points in a community's life, for example, a moderate value may indicate a debate that is still forming or one that is dissolving, and only the trajectory distinguishes the two.

Not all temporal change develops gradually. External events can shift polarization sharply, but the shape of the response depends on the event. Events that are scheduled and widely anticipated can produce a slow build-up beforehand as groups mobilize, whereas sudden and unpredictable events produce abrupt spikes afterward [5]. Such events can also destabilize established alignments, forcing participants to reposition themselves before the community settles again [23]. Temporal analysis therefore has to account for *when* an event occurs relative to the measurement window, not just whether it occurs.

The structural and linguistic dimensions of polarization need not move together. A community's interaction network can remain stably segregated while the language its members use shifts substantially, as the topics, vocabulary, and framing of the discussion are taken up and reshaped by different actors [8]. Conversely, changes in language may precede or fail to produce changes in structure [11]. This temporal decoupling is directly relevant to our study where a measurement derived from text and one derived from network structure may agree in one period and diverge in another, and a comparison between them is only meaningful if it is made across time rather than at a single point.

Several classes of factors have been proposed to explain why polarization moves. At the user level, individual susceptibility to influence and confirmation bias shape how quickly beliefs converge within like-minded groups, and repeated exposure to reinforcing content can narrow what users see over time [25]. At the population level, the arrival of new participants and the departure of intermediate ones alter the composition of the network and can temporarily dilute or sharpen its structure [8, 23]. At the interaction level, the intensity and tone of exchanges between participants carry information about the structure that follows, although the relationship is not always intuitive and can run in unexpected directions [11].

Studying polarization dynamically introduces difficulties that do not arise in static analysis. First, the choice of window size and step involves a trade-off: short windows track rapid shifts but yield sparse graphs, while long windows are stable but blur short-lived changes. Windows with too few interactions give volatile or undefined scores, so minimum-activity thresholds are needed [11]. Second, many structural measures depend on network size and

density, so an apparent change in polarization may simply reflect a change in interaction volume, and so normalization is required to separate genuine change from this artifact [11, 22]. Third, temporal autocorrelation between adjacent windows can cause naive statistical tests to overestimate relationship significance. To account for this temporal dependence, we employ shift-based permutation tests that preserve the internal structure of each time series [11].

These observations motivate evaluating polarization measures over time rather than at a single snapshot. Because both the language-based and the network-based pipelines produce one score per window, we can compare their trajectories against an external reference, asking not only whether each measure is accurate on average but also whether it tracks the formation, reversal, and event-driven shifts described above.

2.5 Research Gaps

LLMs can now measure polarization either by scoring exchanges directly [5] or by converting them into signed networks on which structural measures are computed [11], but the literature leaves three gaps open.

No controlled comparison. The two approaches were introduced and validated separately, on different datasets, time units, and constructs. Neither study tests whether routing text through a network improves on scoring it directly, so differences between the two cannot be attributed to pipeline design unless data, windows, and ground truth are held fixed.

Unknown conditions and sign sensitivity. Structural indices depend on graph size and density [11, 22], and hostility need not align with the main dividing line [9], so neither pipeline should dominate everywhere. Structural estimates are also highly sensitive to edge signs [11], yet how much sign quality (sentiment, binary LLM, continuous LLM) matters for downstream accuracy has not been quantified. t therefore lack guidance on when building a signed network is worth its cost.

Untested temporal value. Polarization is a process [23], and text and structure can move independently [8, 11]. Evaluations typically report agreement in level, which autocorrelation inflates, and no study has tested whether a structure-aware score improves forecasts of *change* in polarization over a text-only score.

Our study addresses these gaps with a controlled, time-resolved comparison of both pipelines against a common external reference.

3 Proposed Approach (Overview)

Our study compares two ways of using an LLM to measure polarization on the same data and against the same external ground truth. Pipeline A (structure-blind) aggregates LLM stance scores directly into a polarization score [5]. Pipeline B (structure-aware) uses the LLM to sign each interaction, builds a confidence-weighted signed network, and measures polarization from its structure [11]. The work is organized into five phases, moving from controlled validation of the measurement tools to the head-to-head comparison and the analysis of when each pipeline wins.

3.1 Phase 1: Controlled Validation

Before applying the proposed pipelines to real-world data, we will first conduct a small controlled validation using synthetic networks with known levels of polarization, together with null models that preserve basic properties such as topology or signed degree. The goal is to check whether the selected polarization measures (e.g., Eigen-Sign and frustration) behave as expected under different levels of structural polarization, whether they distinguish polarized networks from appropriate null models, and whether normalization removes the effect of graph size and density. This phase serves primarily as a sanity check for the measurement methods before moving to real data. All reported findings will come from real, publicly available datasets.

3.2 Phase 2: Real Data and Graph Construction

We will then apply the two pipelines to publicly available datasets containing political or controversial online discussions, selected from the dataset options we discuss in a later sections. The data will be cleaned and organized into comparable interaction units, with users represented as nodes and interactions represented as edges where appropriate. Where timestamps are available, we will split the data into time windows to capture changes in polarization over time. Where topic information is available, we will also treat topics as separate comparison units, so that polarization can be compared across multiple controversial issues rather than a single one. The same underlying data and windows will be used for both pipelines so that their results can be compared directly.

3.3 Phase 3: Core Comparison

The main experimental phase will compare the two proposed approaches to polarization measurement. Pipeline A will estimate polarization directly from the language of interactions, while Pipeline B will use the same language information to construct a signed interaction network and then measure polarization from its structure. We will compare the resulting scores against available external labels or other forms of ground truth. We will also run a small number of baseline comparisons to determine whether the differences between the pipelines are driven by the use of network structure itself or by other aspects of the measurement process. These include comparing simple sentiment-based signs with LLM-inferred signs, binary versus continuous confidence-weighted edges, and a signed network versus the same network with signs ignored.

3.4 Phase 4: Optimization and Extensions

After establishing the main comparison, we will explore possible extensions to assess the robustness and generalizability of the findings. These may include alternative language models, including smaller open models run locally, and alternative edge representations or network-based polarization measures, such as a signed graph neural network [6, 14]. Since real discussions are not always split into exactly two sides, we will also consider multi-camp structure, going beyond two-faction measures to detect more than two opposing groups. If feasible, we will also apply the framework to an additional dataset that falls outside our primary domain to examine whether the findings extend beyond the primary setting.

Table 1: Dataset suite. RQ1: routing through a signed network vs. direct LLM measurement; RQ2: when structure adds value; RQ3: edge-sign accuracy and polarization error; RQ4: forecasting value; RQ5: drivers of change (exploratory). We do not expect every dataset to support every RQ.

Dataset	Text on interactions	User network	Independent ground truth	Time windows	RQs
Reddit Politosphere	Yes (comments)	Yes (reply edges from parent IDs)	Partial (subreddit party tags, proxy)	Yes (monthly, 2008–2019)	1, 2, 4, 5
PolitiSky24	Yes (replies, quotes)	Yes (reply/quote edges; like and repost graphs)	Partial (445 human-labelled users; rest LLM)	Yes, short (Nov. 2022–Nov. 2024)	1, 2
SRQ	Yes (per edge)	No (no user IDs)	Yes (human edge-level stance)	Partial (four 2018 events)	3
Conflict-Bias-Eval	Yes (per comment)	No (no reply links documented)	Yes (human comment-level stance)	Partial (Oct. 2023–Aug. 2024)	1

3.5 Phase 5: Analysis and Contributions

Finally, we will analyze the conditions under which the two approaches produce similar or different assessments of polarization. In particular, we will examine whether network structure provides additional information beyond what can be inferred directly from language, and whether this depends on the characteristics of the discussion, such as topic, network density, or how extreme the language is. Our working hypothesis is that Pipeline B should help most when language is moderate but interaction patterns are established along opposing camps, while Pipeline A should do as well or better when polarization is expressed mainly through charged language. We will also investigate representative cases where the two pipelines disagree to better understand the source of these differences.

The overall goal is to determine whether incorporating interaction structure provides a meaningful advantage for measuring polarization, and to identify the circumstances in which a network-based approach is more informative than measuring polarization directly from text.

4 Datasets

4.1 Requirements

An ideal dataset for our questions would provide (i) text on every interaction, so edge signs can be inferred and text can be scored directly; (ii) a timestamped user-to-user network; (iii) ground truth independent of our LLM pipelines, i.e., human-labelled edge signs (RQ3) and a user- or camp-level polarization reference (RQ1–RQ2); (iv) many comparable units (time windows, communities, topics) with varying structure and enough time points to forecast (RQ2, RQ4); (v) covariates and events for studying drivers of change (RQ5); and (vi) open access. We found no public dataset that provides all six, so we assemble a suite in which each dataset covers different requirements (Table 1).

4.2 Available Datasets

Reddit Politosphere. The Politosphere [12] contains all comments from 605 political subreddits between 2008 and 2019, drawn from the Pushshift Reddit Dataset [2], with pseudonymized authors, plus yearly subreddit-overlap networks and subreddit metadata (including 43 Democratic and 20 Republican party-affiliated subreddits). Comment volume grows from 12.0M in 2015 to 71.0M in 2019. Replies to other comments give user-to-user edges with text, timestamps and parent context. The data is open (CC BY 4.0) at https://doi.org/10.5281/zenodo.5851729 [13], with loaders at https://github.com/valentinhofmann/politosphere. Monthly files are large (roughly 180–570 MB compressed in 2015–2016), so we analyse a subset. *Limitations:* edge signs are not provided, so we infer them; camps are subreddit-level proxies, and party-tagged subreddits are not purely one-sided (the authors note that `r/AskDemocrats` sits near the Republican subreddits); Reddit users skew younger, more male and more liberal; the data ends in 2019. *Relevance:* it is the only dataset here that combines text, a user network and a long (12-year) monthly series, so it carries our main experiments.

PolitiSky24. PolitiSky24 [20] provides user-level stance labels toward Harris and Trump for the 2024 US election on Bluesky: 16,044 user–target pairs for 8,022 users, plus posting histories (18.4M posts from 8,561 users, Nov. 2022–Nov. 2024, with reply, quote, parent-ID and timestamp fields) and like and repost networks. Labels come from a retrieval-plus-LLM pipeline that reaches 81.2% accuracy on a human-labelled validation set of 445 users. The data is CC BY 4.0 (2.1 GB) at https://doi.org/10.5281/zenodo.15616911 [21]. *Limitations:* about 95% of labelled users are LLM-labelled, so they are not independent truth for LLM-based pipelines; the human set is small, hashtag-sampled (possibly favouring extreme users) and one-sided (390 of 445 are against Trump); the cohort was selected from feeds sampled over 16 days in Nov. 2024, post volume surges that month, and labels use original posts and reposts only. We use the human-labelled users as the answer key and do not quote users.

SRQ. SRQ [24] has over 5,200 human-labelled stances (support, denial, comment, query; explicit and implicit) of Twitter replies and quotes toward a target tweet, from three 2018 events plus a general-terms sample. Collection favoured denial-type responses, which the authors consider useful for finding antagonistic relationships. A re-check of 55% of confidently labelled tweets matched in 86.83% of cases. Data: https://doi.org/10.5281/zenodo.3609277. *Limitations:* no user IDs, so no network; only first-level replies; an unnatural class mix; English, Twitter and 2018 only; and stance toward a tweet is not an interpersonal sign. We use it only to measure sign-classifier error against human labels (RQ3). DEBAGREEMENT [19], which has human-labelled agree/disagree edges between Reddit users, would have been the ideal RQ3 resource, but its download is no longer available, so SRQ serves that role.

Conflict-Bias-Eval. Conflict-Bias-Eval [1] has 9,969 Reddit comments on the Israel–Palestine conflict (Oct. 2023–Aug. 2024) labelled Pro-Israel (4,947), Pro-Palestine (2,591) or Neutral (2,431) by three annotators ($\kappa = 0.93$), available at https://github.com/jami78/Conflict-Bias-Eval. We use it as a text-only baseline for direct LLM measurement (RQ1). *Limitations:* comments were selected by polarizing keywords and annotation was keyword-guided; the paper documents no reply links, so there is no network; it covers one topic with a random train/test split; no licence is shown on the repository.

4.3 Coverage of the Research Questions

Politosphere is the main testbed for RQ1–RQ2 and the only long series for RQ4–RQ5. PolitiSky24 repeats RQ1–RQ2 on a second platform with human-labelled users. SRQ supplies measured sign-error rates, which we inject as noise into the networks to study RQ3. Conflict-Bias-Eval gives the direct-measurement baseline for RQ1. We do not expect every dataset to support every RQ, and given project time and the limitations above, we may not address all five RQs in full.

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References

[1] Hasin Jawad Ali, Ajwad Abrar, S. M. Hozaifa Hossain, and M. Firoz Mridha. 2025. Social media polarization during conflict: Insights from an ideological stance dataset on Israel-Palestine Reddit comments. https://doi.org/10.48550/arXiv.2502.00414 arXiv:2502.00414 [cs.CL]

[2] Jason Baumgartner, Savvas Zannettou, Brian Keegan, Megan Squire, and Jeremy Blackburn. 2020. The Pushshift Reddit Dataset. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 14. 830–839. https://doi.org/10.1609/icwsm.v14i1.7347

[3] Aaron Bramson, Patrick Grim, Daniel J. Singer, William J. Berger, Graham Sack, Steven Fisher, Carissa Flocken, and Bennett Holman. 2017. Understanding Polarization: Meanings, Measures, and Model Evaluation. *Philosophy of Science* 84, 1 (2017), 115–159.

[4] Dorwin Cartwright and Frank Harary. 1956. Structural balance: a generalization of Heider's theory. *Psychological Review* 63, 5 (1956), 277–293. https://doi.org/10.1037/h0046049

[5] Jawad Chowdhury, Rezaur Rashid, and Gabriel Terejanu. 2026. Measuring Social Media Polarization Using Large Language Models and Heuristic Rules. arXiv:2601.00927 [cs.SI]

[6] Tyler Derr, Yao Ma, and Jiliang Tang. 2018. Signed Graph Convolutional Networks. In *2018 IEEE International Conference on Data Mining (ICDM)*. 929–934. https://doi.org/10.1109/ICDM.2018.00113

[7] Edoardo Di Martino, Matteo Cinelli, Roy Cerqueti, and Walter Quattrociocchi. 2025. Quantifying Polarization: A Comparative Study of Measures and Methods. arXiv:2501.07473 [cs.CY]

[8] Kevin Durrheim and Maria Schuld. 2025. Polarization on social media: Comparing the dynamics of interaction networks and language-based opinion distributions. *Political Psychology* 46, 6 (2025), 1601–1617. https://doi.org/10.1111/pops.70000

[9] Emma Fraxanet, Max Pellert, Simon Schweighofer, Vicenç Gómez, and David Garcia. 2024. Unpacking polarization: Antagonism and alignment in signed networks of online interaction. *PNAS Nexus* 3, 12 (2024), pgae276. https://doi.org/10.1093/pnasnexus/pgae276

[10] Max Griswold, Michael W. Robbins, and Michael S. Pollard. 2026. Stay Tuned: Improving Sentiment Analysis and Stance Detection Using Large Language Models. *Political Analysis* 34, 3 (2026), 392–411. https://doi.org/10.1017/pan.2025.10023

[11] Zhijin Guo, Li Zhang, Tyler Bonnet, Janet B. Pierrehumbert, and Xiaowen Dong. 2026. Linking Scalar-Intensity Language to Structural Polarization with Validated Signed-Network Measures. arXiv:2605.12814 [cs.SI] Version 1 was titled "Linking Extreme Discourse to Structural Polarization in Signed Interaction Networks".

[12] Valentin Hofmann, Hinrich Schütze, and Janet B. Pierrehumbert. 2022. The Reddit Politosphere: A Large-Scale Text and Network Resource of Online Political Discourse. In *Proceedings of the International AAAI Conference on Web and Social Media*, Vol. 16. 1259–1267. https://doi.org/10.1609/icwsm.v16i1.19377

[13] Valentin Hofmann, Hinrich Schütze, and Janet B. Pierrehumbert. 2022. The Reddit Politosphere (dataset). https://doi.org/10.5281/zenodo.5851729

[14] Junjie Huang, Huawei Shen, Liang Hou, and Xueqi Cheng. 2021. SDGNN: Learning Node Representation for Signed Directed Networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 196–203. https://doi.org/10.1609/aaai.v35i1.16093

[15] Shanto Iyengar, Yphtach Lelkes, Matthew Levendusky, Neil Malhotra, and Sean J. Westwood. 2019. The Origins and Consequences of Affective Polarization in the United States. *Annual Review of Political Science* 22 (2019), 129–146. https://doi.org/10.1146/annurev-polisci-051117-073034

[16] Shanto Iyengar, Gaurav Sood, and Yphtach Lelkes. 2012. Affect, Not Ideology: A Social Identity Perspective on Polarization. *Public Opinion Quarterly* 76, 3 (2012), 405–431. https://doi.org/10.1093/poq/nfs038

[17] Neta Kligler-Vilenchik, Christian Baden, and Moran Yarchi. 2020. Interpretative Polarization across Platforms: How Political Disagreement Develops over Time on Facebook, Twitter, and WhatsApp. *Social Media + Society* 6, 3 (2020), 2056305120944393. https://doi.org/10.1177/2056305120944393

[18] Emily Kubin and Christian von Sikorski. 2021. The role of (social) media in political polarization: a systematic review. *Annals of the International Communication Association* 45, 3 (2021), 188–206. https://doi.org/10.1080/23808985.2021.1976070

[19] John Pougué-Biyong, Valentina Semenova, Alexandre Matton, Rachel Han, Aerin Kim, Renaud Lambiotte, and Doyne Farmer. 2021. DEBAGREEMENT: A comment-reply dataset for (dis)agreement detection in online debates. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*. https://openreview.net/forum?id=udVUN__gFO

[20] Peyman Rostami, Vahid Rahimzadeh, Ali Adibi, and Azadeh Shakery. 2025. PolitiSky24: U.S. Political Bluesky Dataset with User Stance Labels. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, Suzhou, China, 21976–21993. https://doi.org/10.18653/v1/2025.findings-emnlp.1198

[21] Peyman Rostami, Vahid Rahimzadeh, Ali Adibi, and Azadeh Shakery. 2025. POLITISKY24: U.S. Political Bluesky Dataset with User Stance Labels (dataset). https://doi.org/10.5281/zenodo.15616911

[22] Ali Salloum, Ted Hsuan Yun Chen, and Mikko Kivelä. 2022. Separating Polarization from Noise: Comparison and Normalization of Structural Polarization Measures. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW1 (2022), 1–33. https://doi.org/10.1145/3512962

[23] Célina Treuillier, Sylvain Castagnos, Christèle Lagier, and Armelle Brun. 2024. Gaining a better understanding of online polarization by approaching it as a dynamic process. *Scientific Reports* 14 (2024), 8702. https://doi.org/10.1038/s41598-024-58915-w

[24] Ramon Villa-Cox, Sumeet Kumar, Matthew Babcock, and Kathleen M. Carley. 2020. Stance in Replies and Quotes (SRQ): A New Dataset For Learning Stance in Twitter Conversations. https://doi.org/10.48550/arXiv.2006.00691 arXiv:2006.00691 [cs.CL]

[25] Xiaodan Wang, Yanbin Liu, Shiqing Wu, Ziying Zhao, Yuxuan Hu, Weihua Li, and Quan Bai. 2026. Ideological Isolation in Online Social Networks: A Survey of Computational Definitions, Metrics, and Mitigation Strategies. arXiv:2601.07884 [cs.SI]